Hair 3

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Consultant dermatology

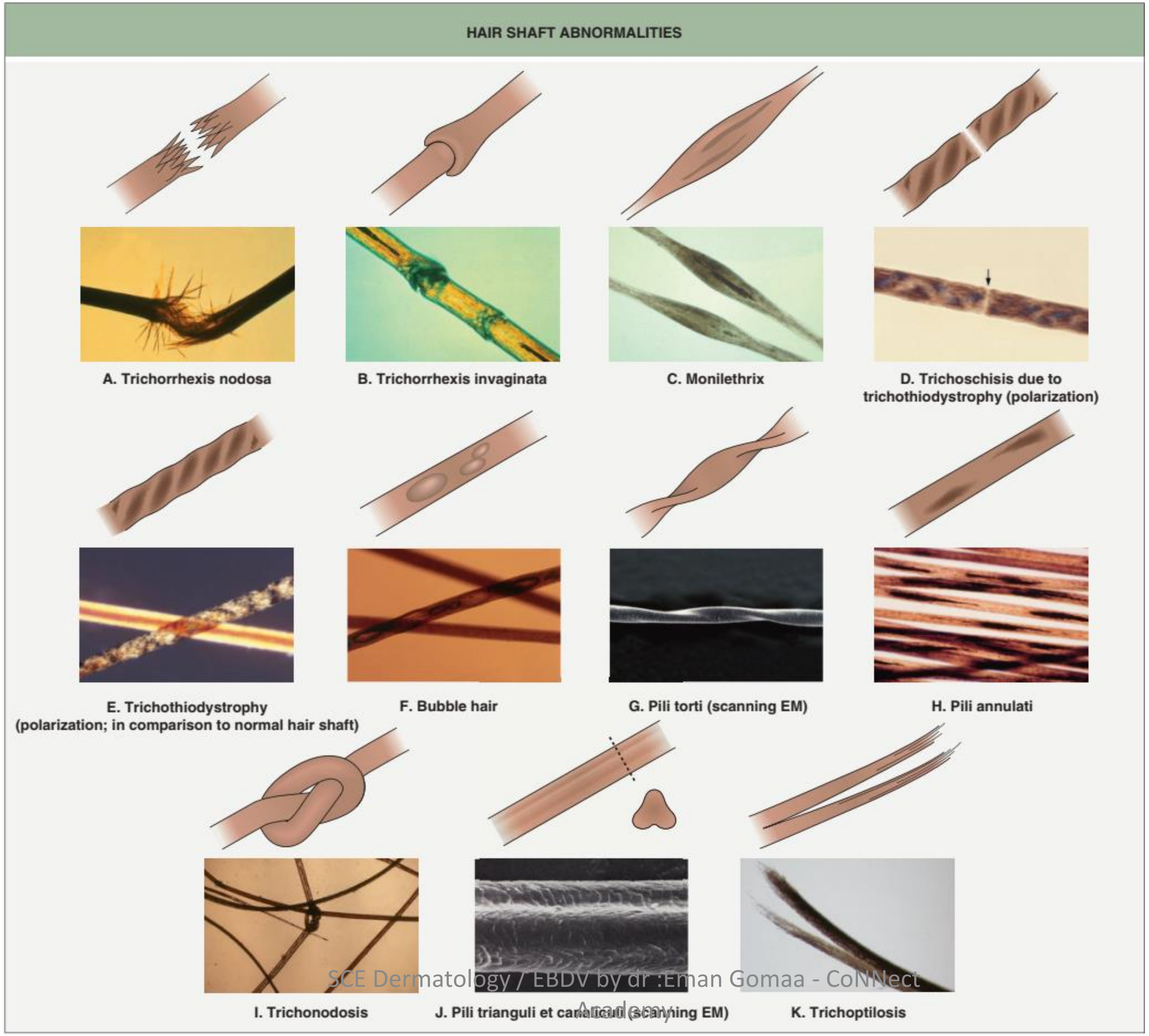

HAIR SHAFT ABNORMALITIES
A. Trichorrhexis nodosa
B. Trichorrhexis invaginata
C. Monilethrix
D. Trichoschisis due to trichothiodystrophy (polarization)
E. Trichothiodystrophy (polarization; in comparison to normal hair shaft)
F. Bubble hair
G. Pili torti (scanning EM)
H. Pili annulati
I. Trichonodosis
J. Pili trianguli et canaliculi (scanning EM)
K. Trichoptilosis

Fig. 69.26 Monilethrix. Hypotrichosis due to breakage of fragile hairs. Note the small perifollicular papules with

Fig. 69.27 Woolly hair nevus. Circumscribed area where scalp hair is wavier and has a woolly appearance.

- Disorder of sweat gland
  - Eccrine
    - ypo/jyperhidrosis
    - Neutophilicecrin hidradenitis
    - Miliaria
  - Apocrine
    - Hidradenitis suppurativa
    - Fox Fordyc

Hyperhidrosis

	1ry hyperhidrosis	2ry hyperhidrosis
	Equal focal Localized → palm, sole, forehead & axilla.	Localized & generalized
causes	Familial, AD 60-80 % Emotional stress	Neural → cortical [emotional], hypothalamic [thermal], medullary [gustatory], spinal [abnormal segmental sweating], autonomic & compensatory. Non-neural → direct stimulus → heat, drugs → dopamine, epinephrine, pilocarpine, antipyretic & opioid
diagnostic	Diagnostic criteria → focal, no cause, > 6 months, +ve family history, bilateral & > 25 ys	
inv		
ttt	Aluminum chlorid, botox Drug → oxibutinin [uripan] Iontophoresis Surgical	

CRITERIA FOR THE DIAGNOSIS OF PRIMARY HYPERHIDROSIS
1. Focal, visible excess sweating 2. Present for at least 6 months 3. No apparent secondary causes 4. At least two of the following: • Bilateral and symmetric • Impairs activities of daily life • At least one episode per week • Age of onset <25 years • Positive family history • Stops during sleep

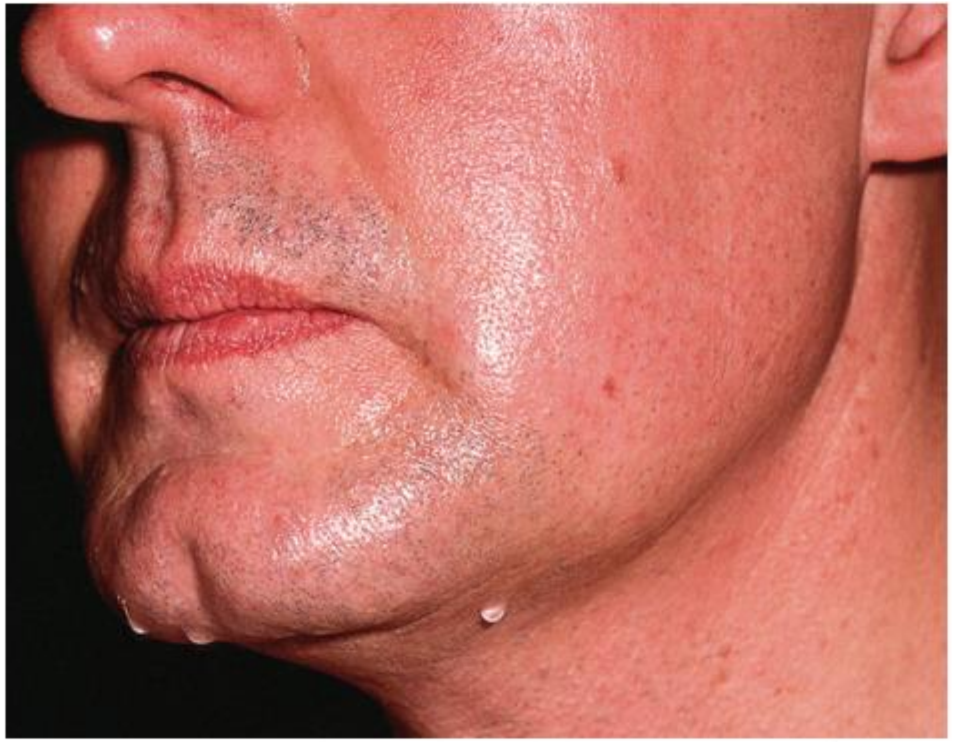


Fig. 39.1 Primary cortical (emotional) hyperhidrosis involving the face. Sweat droplets are evident on the upper cutaneous lip, jawline, and chin. *Reproduced*

DRUGS THAT CAN STIMULATE ECCRINE SWEATING	
Direct-acting cholinomimetic agents	
• Acetylcholine	• Anti-xerostomia drugs (see Table 45.5)
Cholinesterase inhibitors	
• Anti-Alzheimer drugs* • Anticholinergic antidote	• Antimyasthenics
Adrenomimetic agents	
• Dopamine • Epinephrine (adrenaline) • Isoproterenol	• Norepinephrine (noradrenaline) • Phenylpropanolamine
Antidiabetic (hypoglycemic) agents	
• Insulins	• Sulfonylureas
CNS stimulants	
• Amphetamines • Caffeine	• Theophylline
Antidepressants	
• MAOIs† • SSRIs†	• Tricyclics†
Antipsychotics	
Antipyretics	
Opioids	
Other drugs	
• ACE inhibitors • Amlodipine • Atomoxetine	• Phosphodiesterase inhibitors (e.g. sildenafil) • Sibutramine†

CAUSES OF CORTICAL (STRESS/EMOTIONAL) HYPERHIDROSIS
Primary hyperhidrosis
Secondary hyperhidrosis
Disorders of cornification
• Palmoplantar keratodermas • Pachyonychia congenita • Congenital ichthyosiform erythroderma and epidermolytic ichthyosis
Other genodermatoses
• Epidermolysis bullosa: simplex > junctional • Dermatopathia pigmentosa reticularis • Dyskeratosis congenita • Pachydermoperiostosis (digital clubbing and thick, furrowed skin on the face and scalp) • Apert syndrome (craniosynostosis, digital anomalies, severe acne; *FGFR2* mutations) • Nail–patella syndrome
Hereditary sensory and autonomic neuropathies (HSANs)[†]
• Familial dysautonomia (Riley–Day syndrome; HSAN type III; blotchy erythema accompanies episodic hyperhidrosis of the face and trunk, decreased lacrimation, postural hypotension, and dysautonomic crises [vomiting, hypertension]) • Congenital autonomic dysfunction with universal pain loss • Congenital sensory neuropathy (HSAN type II)

Fig. 39.3 Gustatory sweating in the auriculotemporal (Frey) syndrome, as a consequence of parotid surgery. The blue–black area represents sweating (starch–iodine technique). Salivary stimulation induced this sweating response.

CAUSES OF HYPOTHALAMIC HYPERHIDROSIS						
Infection	Tumors	Endocrine/ metabolic	Vasomotor	Neurologic	Drugs/toxins	Miscellaneous
• Acute febrile illnesses (defervescence) • Tuberculosis • Brucellosis • Malaria • Subacute bacterial endocarditis • Other	• Lymphoma • Pheochromocytoma* • Carcinoid*	• Hyperthyroidism • Acromegaly • Hypoglycemia • Obesity • Menopause • Pregnancy • Porphyria (acute intermittent) • Phenylketonuria • Gout	• Congestive heart failure† • Myocardial ischemia • Raynaud phenomenon • Acrocyanosis • Cold injury • Symmetrical lividity of the palms and soles‡ • Reflex sympathetic dystrophy • Autoimmune connective tissue disease (e.g. rheumatoid arthritis, SLE)	• CNS tumors • Cerebrovascular accidents • Parkinson disease • Postencephalitic • Episodic spontaneous hypothermia with hyperhidrosis§ • Cold-induced sweating syndrome‖ • Familial dysautonomia‡	• Alcoholism • Alcohol or opioid withdrawal • Mercury toxicity (acrodynia) • Chronic arsenic toxicity • See Table 39.2	• Compensatory hyperhidrosis (e.g. in association with miliaria, diabetic neuropathy, sympathectomy)¶ • Localized unilateral hyperhidrosis (face and neck, forearm)‡ • POEMS syndrome • Speckled lentiginous nevus syndrome • Mitochondrial disorders • Pressure and postural hyperhidrosis‡

OTHER CAUSES OF SECONDARY HYPERHIDROSIS			
Medullary (gustatory)	**Spinal cord**	**Axon reflex**	**Abnormal blood vessels or eccrine glands (non-neural)**
• Physiologic gustatory sweating • Frey syndrome • Chorda tympani syndrome • Cluster headaches • Sympathectomy • Sympathetic trunk and vagus nerve injury (e.g. by lung carcinoma, mesothelioma, thyroidectomy, subclavian aneurysm) • Encephalitis • Syringomyelia	• Spinal injury • Syringomyelia • Tabes dorsalis	• Inflammatory skin diseases • Drugs (see Table 39.2)	• Aquagenic palmoplantar keratoderma • Eccrine nevi • Eccrine angiomatous hamartoma (Fig. 39.5) • Maffucci syndrome • Arteriovenous fistula • Klippel–Trenaunay syndrome • Glomuvenous malformation • Blue rubber bleb nevus syndrome • Cold erythema • Granulosis rubra nasi • Pretibial myxedema

INITIAL PATIENT EVALUATION FOR SECONDARY HYPERHIDROSIS	
Laboratory test	**Disease**
Serum electrolytes, BUN, creatinine	Renal disease (rare)
Fasting blood glucose level and/or HbA1c	Diabetes mellitus
Thyroid function tests	Hyperthyroidism
Skin and/or blood tests for tuberculosis (e.g. PPD, QFTB-G)	Tuberculosis
Chest X-ray	Tuberculosis, neoplasm
Complete blood count	Infection
Sedimentation rate/C-reactive protein test	Infection, neoplasm, inflammatory disease
Antinuclear antibodies	Autoimmune connective tissue disease
Urinary catecholamines*	Pheochromocytoma

eFig. 39.1 Sweating pattern in axillary hyperhidrosis. Blue–black areas represent foci of sweating, as demonstrated by the starch–iodine technique: iodine solution (e.g. 3.5% in alcohol) is applied to clean, shaved skin and allowed to dry, then starch powder (e.g. cornstarch) is brushed onto the area.

TREATMENT OF HYPERHIDROSIS					
	Therapy	**Frequency & recommended dose**	**Side effects**	**Duration**	**Comments**
First-line	*Topicals* Aluminum chloride hexahydrate – 6.25 to 20% Aluminum chloride – 12% to 20% Aluminum zirconium salts	Use nightly for 3–5 nights, then every few days as needed	Burning Irritant contact dermatitis	Days	Blocks sweat ducts Aluminum zirconium salts may be effective in axillae, but not volar surfaces
Second-line	*Acetylcholine release inhibitors, i.e. botulinum toxins*	Every 4–6 months	Discomfort during injection Weakness of underlying muscles	Months	Prevents release of acetylcholine
	Iontophoresis	2–3 times a week	Discomfort during procedure	Days	Blocks sweat ducts
	*Oral therapy**	As needed		Hours	
	Oxybutynin	1.25–5 mg BID	Dry mouth and urinary retention most common; also confusion and decreased mental status		Anticholinergic
	Glycopyrrolate	1–2 mg BID			Anticholinergic
	Clonidine	0.1–0.3 mg BID	Hypotension, rebound hypertension		α_2-Adrenergic agonist
	Propranolol	10–40 mg BID	Hypotension, bradycardia, hyperhidrosis with long-term use		β-Adrenergic blocker
	Clonazepam	0.25–0.5 mg BID	Sedation		Anxiolytic
	Thermal ablation Nd:YAG laser Microwave	Multiple	Expensive	Years	
Third-line	*Liposuction*	Once to more than once**	Expensive	Semipermanent to permanent	
	Local excision	Once	Scarring	Permanent	Last resort
	Sympathectomy	Once	Compensatory hyperhidrosis Horner syndrome	Usually permanent	

hypohydrosis

Central	Peripheral	idiopathic
Tumor → hypothalamus, medulla & spinal cord Horner syndrome Shydragor syndrome Ross syndrome Drugs	Genetic → ectodermal dysplasia, IP Destruction → tumor & burn Obstruction → miliaria & ichthyosis	

CAUSES OF CENTRAL AND NEUROPATHIC HYPOHIDROSIS AND ANHIDROSIS
• Tumors, infarctions, and other lesions of the hypothalamus, pons or medulla • Spinal cord tumors and injuries • Horner syndrome • Degenerative syndromes - Pure autonomic failure - Multiple system atrophy (Shy–Drager syndrome)* - Ross syndrome (see text) • Autoimmune autonomic neuropathy • Congenital insensitivity to pain with anhidrosis • Peripheral neuropathy due to: - Diabetes - Alcoholism - Leprosy - Amyloidosis (immunoglobulin light chain and transthyretin types) • Drugs (see Table 39.9)

DRUGS THAT CAUSE HYPOHIDROSIS AND ANHIDROSIS	
Blockade of neurotransmission	
• **Nicotinic acetylcholine receptor antagonists (ganglion-blocking agents)**	
- Hexamethonium	- Tetraethylammonium
- Mecamylamine	- Trimethaphan
• **Muscarinic acetylcholine receptor antagonists***	
- Atropine	- Oxybutynin
- Glycopyrrolate	- Scopolamine
• **Calcium channel blockers**	
• **α-Adrenergic blockers**	
- Phenoxybenzamine (hypohidrosis occasionally observed)	
- Phentolamine†	
• **α_2-Adrenergic agonists‡**	
- Clonidine	
Dysfunction or destruction of the eccrine gland§	
• Aldehydes (topical)	• Topiramate¶
• Aluminum salts (topical)	• Zirconium salts (topical)
• 5-Fluorouracil	• Zonisamide¶
• Quinacrine	

PERIPHERAL CAUSES OF HYPOHIDROSIS AND ANHIDROSIS DUE TO SWEAT GLAND ABNORMALITIES
Genetic disorders
• Ectodermal dysplasias - Hypohidrotic ectodermal dysplasia - Hypohidrotic ectodermal dysplasia with immune deficiency - Rapp–Hodgkin syndrome - Naegeli–Franceschetti–Jadassohn syndrome • Incontinentia pigmenti • Bazex–Dupré–Christol syndrome • Fabry disease*
Destruction
• Tumors (e.g. lymphoma) • Burns • Radiation therapy • Systemic sclerosis (scleroderma) and morphea • Sjögren syndrome† • Graft-versus-host disease • Acrodermatitis chronica atrophicans
Obstruction
• Miliaria • Ichthyoses (e.g. lamellar ichthyosis) • Psoriasis • Eczematous dermatoses • Bullous diseases (e.g. epidermolysis bullosa [inherited or acquired]) • Porokeratosis

miliaria

- Obstruction of ecrine duct → sweat retention & inflammation.
- Common in children
- Overheating
- > in forehead & trunk
- Complicated by → hyperpyrexia & lymphadenopathy.

3 types of miliaria

Miliaria crystalina	Miliaria rubra	Miliaria profunda
• Superficial. • Neonatal. • Upper trunk & face. • asymptomatic	Intermediate. Neonatal. Neck & upper trunk. Pruritic.	Deep. Adult. Trunk & extrimities. pruritic

Fig. 39.8 Miliaria crystallina. Multiple small superficial ves with clear fluid.

THREE TYPES OF MILIARIA				
Type	Location of obstruction	Cutaneous lesions	Patient population(s)	Most common locations
Crystallina	Stratum corneum	Non-pruritic, clear, fragile, 1 mm vesicles	Neonates <2 weeks of age Children and adults in hot climates	Face and trunk (Fig. 39.8)
Rubra	Mid-epidermis	Pruritic, erythematous, 1–3 mm papules; may have pustules (miliaria pustulosa)	Neonates 1–3 weeks of age Children and adults in hot climates	Neck and upper trunk (Fig. 39.9)
Profunda	Dermal–epidermal junction	Non-pruritic, white, 1–3 mm papules	Adults in hot climates; often with multiple bouts of miliaria rubra	Trunk and proximal extremities

TTT

- Cooling of environment.
- Topical steroid & AB

Neutophilic ecrine hyrdadenitis [chemotherapy] ecrine hydradenitis

- Child & adult male.
- Cytrabin – anthracycline
- Cytotoxic process → chemotherapy is excreted in sweat gland → destruction
- c/pic → painful erythematous papules & plaques → polymorphic & pruritic.
- Febrile neutropenic
- Treated by steroid – dapson & ab

Fig. 39.10 Neutrophilic eccrine hidradenitis. Erythematous plaques on the

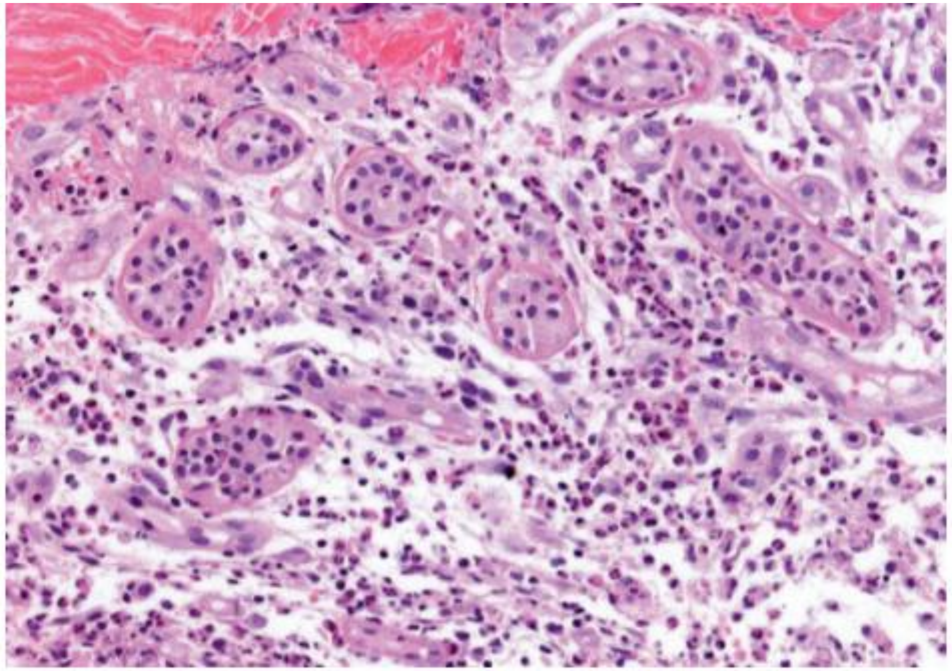


Fig. 39.11 Neutrophilic eccrine hidradenitis – histologic features. Neutrophils surround and infiltrate the eccrine glands and coils. There are degenerative

Fig. 39.12 Idiopathic palmoplantar hidradenitis. Erythematous papulonodules on the plantar surface. *Courtesy, Michael L Smith, MD.*

Fig. 39.13 Keratolyis exfoliativa. Small annular collarettes of scale on the palm. *Courtesy, Jean L Bolognia, MD.*

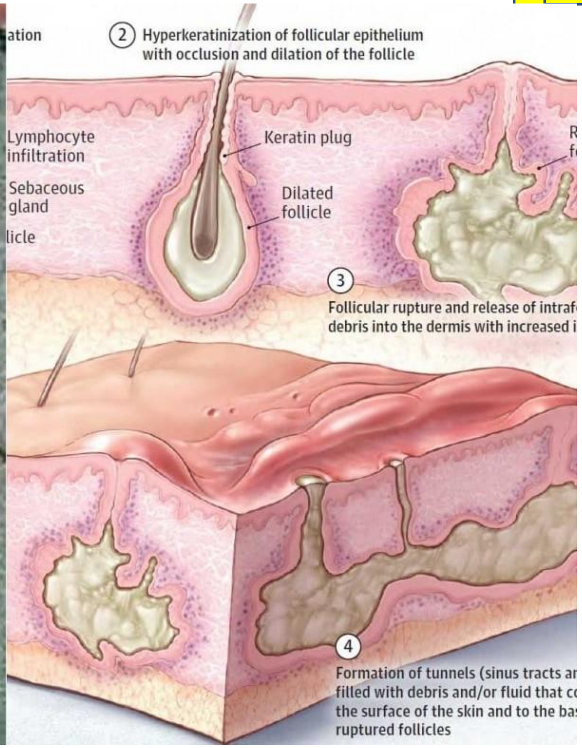
Hydradenitis suppurativa

- Chronic inflammatory disorder in apocrine gland bearing skin [hair follicle].
- Axilla, groin, anogenital & inframammary
- Puberty, women 3:1 & african

aetiopathogenesis

- Occlusion of hair follicle → rupture → release Content → inflammation → destruction → Sinus tract.

Follicular occlusion →
not infection
Not apocrine gland



Genetic
Familial AD

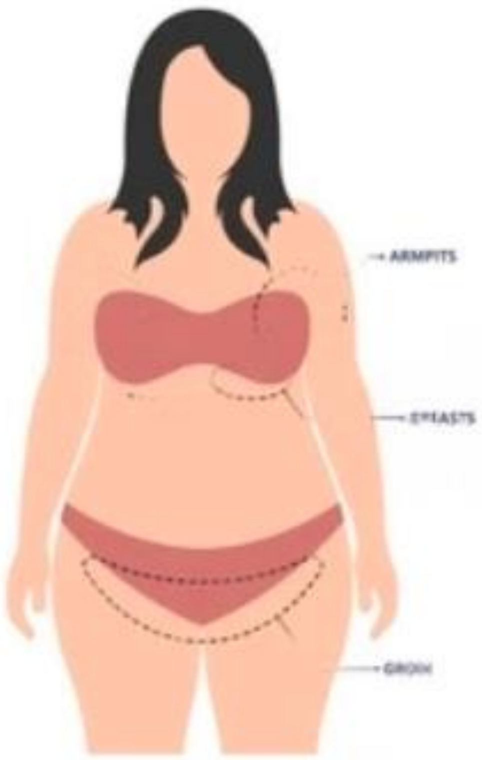
Environmental
Smoking, obesity & lithium

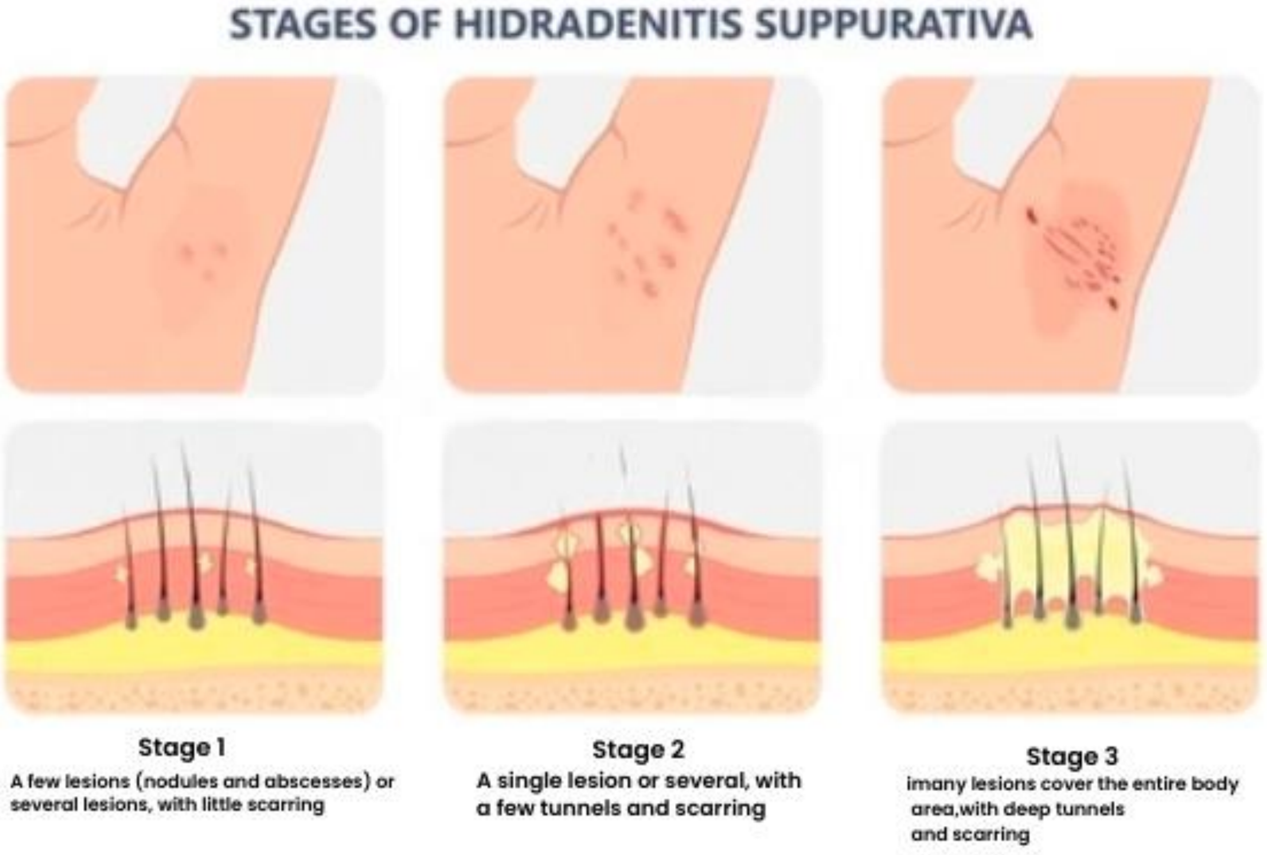
Immunological
Innate, acquired & IL 1 beta

c/pic

- Very tender inflammatory nodules & sterile abscess → sinus tract → douple ended pseudo commedon → discharge serious blood & pus.→ hypertrophic scars.
- **EXTREMELY PAINFUL**

HARLY STAGING SYSTEM





H.P. → rarely needed

- Heavy mixed inflammatory infiltrate in lower dermis
- DD →
- Complications →
    1. Psychological.
    2. Cutaneous → lymphedema, fistula & SCC.
    3. Systemic.

TTT

1. General measures.
2. Topical
    - clindamycin
    - mupricin

3. Systemic
    - rifampicin
    - clindamycin
    - retenoids
    - cyclosporins
    - biologics?

4.Surgical

5.Laser.

6.others

Fox fordyce [apocrine miliaria]

- Chronic pruritic papule → occlusion of apocrine gland.
- Axilla
- Female puberty, hormonal, androgen, pregnancy & OCP.
- Ttt → topical steroid, keratolytic & anti-androgen

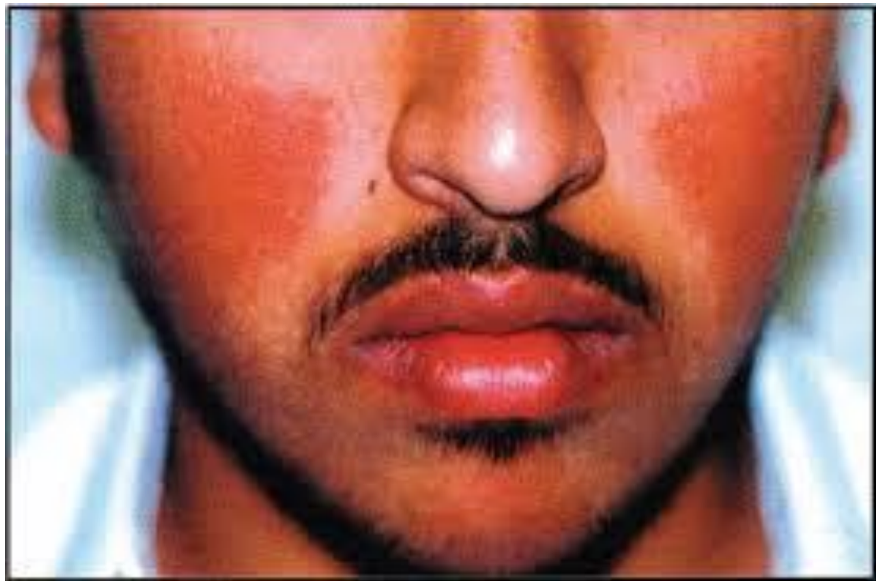
Fig. 39.14 Fox–Fordyce disease.
A,B Monomorphic skin-colored papules in the axillary vault. *Courtesy, Jean L Bolognia, MD.*

Disorder of follicular keratinization

Erythromelanosis follicularis fascii et coli
Keratosis pilaris atrophicans
Lichen spinolosus
Keratosis pilaris
Phirynoderma
LPP
PRP

Erythromelanosis follicularis fascii et coli

- Triad → tiny follicular papule, erythema & hyperpigmentation in cheeks & lateral neck



Keratosis pilaris atrophicans

1. Ulerythema ophryogenesis.
2. Atrophoderma vermiculatum.
3. Keratosis follicularis spinolosa decalvan

Lichen spinolosus

- Nutmeg grater.
- Neck & shoulder.
- HIV infection !!!!!!!!!!!!

Figure 2.

Keratosis pilaris

- V. common 50-80 %.
- Variant of normal skin rather than disease

Phirynoderma

- Vit A def.
- Toed skin
- Conical keratotic plug

Hyperpigmentation disorder

A → Facial melanosis

1] melasma

2] poikiloderma of Sivatte

3] peribuccal pigmentation of Broq

4] erythromelanosis follicularis.

5] Rheils melanosis

6] Hori nevus.

B → LPP.

C → achy dermatosis

D → PIH

E → Drug induced

Linear hyperpigmentation
Linear PIH
Flagillated pigmented of pleomycin
Hyperpigmentation along plashko

Reticulated hyperpigmentation

DISORDERS CHARACTERIZED BY RETICULATED PIGMENTATION	
Disorder	**Key features**
Confluent and reticulated papillomatosis	Elevated with rough texture; favors neck and upper trunk (see Fig. 67.20); responds well to minocycline
Erythema ab igne	Widely spaced net that corresponds to vascular pattern; due to heat injury (see Ch. 88)
Atopic dirty neck	Anterolateral neck
Prurigo pigmentosa	Typically young Japanese women; favors the back, neck, and chest; recurrent crops of pruritic papulovesicles that resolve with a reticulated pattern of hyperpigmentation
Dyskeratosis congenita*	XR > AD and AR forms (see Table 67.8); pterygium/nail dystrophy (see Fig. 67.23C), leukoplakia, pancytopenia; increased risk of mucosal squamous cell carcinoma and leukemia
Naegeli–Franceschetti–Jadassohn syndrome	AD due to *KRT14* mutations; fading reticulated hyperpigmentation, dental anomalies, hypohidrosis, palmoplantar keratoderma, hypoplastic dermatoglyphs, nail dystrophy
Dermatopathia pigmentosa reticularis	AD due to *KRT14* mutations; persistent reticulated hyperpigmentation (see Fig. 67.24), alopecia, nail dystrophy; palmoplantar keratoderma and hypoplastic dermatoglyphs in some patients
X-linked reticulate pigmentary disorder	XR due to *POLA1* mutations; generalized reticulated pigmentation in male patients (see Fig. 67.25); neonatal colitis, recurrent pneumonia, hypohidrosis, photophobia; amyloid deposits in adults
Dowling–Degos disease (DDD)	AD due to mutations in *KRT5*, *POFUT1*, and *POGLUT1* genes; reticulated hyperpigmentation favoring major flexures (see Fig. 67.26); may be composed of or admixed with discrete hyperpigmented macules and papules; variant associated with hidradenitis suppurativa due to *PSENEN* mutations
Galli–Galli disease	Acantholytic variant of DDD
Haber syndrome	Rosacea-like facial eruption plus the clinical features of DDD
Pigmentatio reticularis faciei et colli	Possible variant of DDD; hyperpigmentation of the face and neck plus multiple epidermoid cysts
Reticulate acropigmentation of Kitamura	AD due to *ADAM10* mutations; atrophic acral lentigo-like lesions, palmoplantar and dorsal phalangeal pitting
Epidermolysis bullosa simplex (EBS) with mottled pigmentation* and generalized severe EBS	AD due to *KRT5/14* mutations, with a specific P25L *KRT5* mutation usually underlying EBS with mottled pigmentation (see Ch. 32)
Mendes da Costa disease*	Traumatic bullae and hyperpigmentation, usually acral; dwarfism, atrichia
Cantu syndrome	Reticulated pigmentation of face, forearms and feet; palmoplantar keratoderma
Reticulate genital hyperpigmentation associated with localized vitiligo*	Congenital or acquired reticulate hyperpigmentation of the genitalia, followed by the development of vitiligo in the same region
Mitochondrial disorders	Sun-exposed areas (e.g. cheeks, dorsal aspects of the forearms and hands); additional cutaneous findings include hypertrichosis, hair shaft abnormalities and acrocyanosis (see Ch. 63)
Postinflammatory	Lichenoid dermatoses (e.g. lichen planus pigmentosus); contact dermatitis (e.g. due to benzoyl peroxide)
Drug-induced	Diltiazem (sun-exposed areas); chemotherapeutic agents such as 5-fluorouracil and bleomycin (see Table 67.4)

Melasma

- Common acquired circumscribed hyperpigmentation of the face.
- Unknown
- Exacerbating factors

1. Sun exposure.
2. Hormonal → pregnancy & OCP.

- Hyperfunctioning melanocyte

Fig. 67.5 Various forms of melasma and melasma-like hyperpigmentation. A Malar variant. **B** Mild centrofacial type with sparing of the philtrum. **C** Extension of the hyperpigmentation onto the mandible. **D** Involvement of the extensor forearm; note the same irregular outline as is seen on the face. **E** Melasma-like appearance in a patient with previous acute cutaneous lupus erythematosus. *C–E, Courtesy, Jean L Bolognia, MD.*

c/pic

- Face, upper chest & extesnors.
- Symmetric patch of hyperpigmentation with irreg. border.
- 3 classic pattern

- 3 classic pattern

1- central face.

2- malar.

3- mandibular

4 types

1- epidermal.
2- dermal
3- mixed
4- indeterminate

inv

- Wood's light.
- H.P.

ttt

- General measure.
- Active ttt

1st line → kligman formula

2nd line → chemical peeling

TREATMENT OPTIONS FOR MELASMA

Recommendations for all patients

- Avoidance of sun exposure and tanning beds
- Daily use of broad-spectrum sunscreen (ideally SPF ≥30 with physical blocker such as zinc oxide or titanium dioxide)
- Sun-protective hats and clothing
- Camouflage makeup
- Discontinue oral contraceptives, if possible

Active treatment*,**

First-line topical therapies

- Triple combination of HQ + retinoid + corticosteroid§ at bedtime
- 4% HQ daily, typically at bedtime
- Azelaic acid (15–20%)

Adjunctive topical therapies

- L-ascorbic acid (10–15%)
- Kojic acid (1–4%)

Second-line therapies

- Glycolic (start at 30% and increase as tolerated) or salicylic acid peels (20–30%) every 4–6 weeks

Third-line therapies¶

- Fractional laser
- Intense pulsed light (IPL)